A fair start.

80
adults



HKD
Healthy ketogenic diet
41

ERD
Energy-restricted diet
39

Stratified by sex + BMI

Concealed allocation
Third party · opaque envelopes

Lim et al., Nutrients 2024;16:4380 · Methods 2.4 · doi:10.3390/nu16244380

Same support. Measured outcomes.

HKD

- Workshops
- App + dietitian

=

ERD

- Workshops
- App + dietitian

kg

Calibrated scale

Primary outcome: weight change at 6 months

Lim et al. 2024 · Methods 2.5–2.7 · Shared support improves the fairness of the programme comparison.

The calorie gap.

HKD

−704

kcal/day

ERD

−423

kcal/day

Reported adjusted difference in change

Both groups reduced intake; HKD reduced it more.

−244

kcal/day

Lim et al. 2024 · Table 5 · Self-reported energy intake; −244 is the adjusted contrast, not −704 minus −423.

Two things changed.



Diet composition

Lower energy intake

Possible mediator



More weight loss

Which pathway explains the difference?

IMPROVEMENT

Isocaloric controlled feeding

Match calories. Test composition.

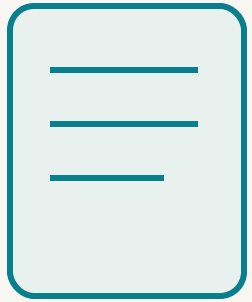
Lim et al. 2024 · Table 5 + Discussion · Possible pathways shown; mediation was not established.

Assigned is not verified.

THE PRESCRIPTION

≤50 g

net carbohydrate/day



Food diaries



Ketosis?

No objective ketone verification reported.

Self-reported intake cannot confirm sustained nutritional ketosis.

Lim et al. 2024 · Methods 2.3, 2.7 + Discussion · No measurement does not mean ketosis never occurred.

Measure the missing link.



Measure ketones

Blood or urine

Repeated checks



Report attainment

Carb target met?

Ketosis confirmed?



Track over time

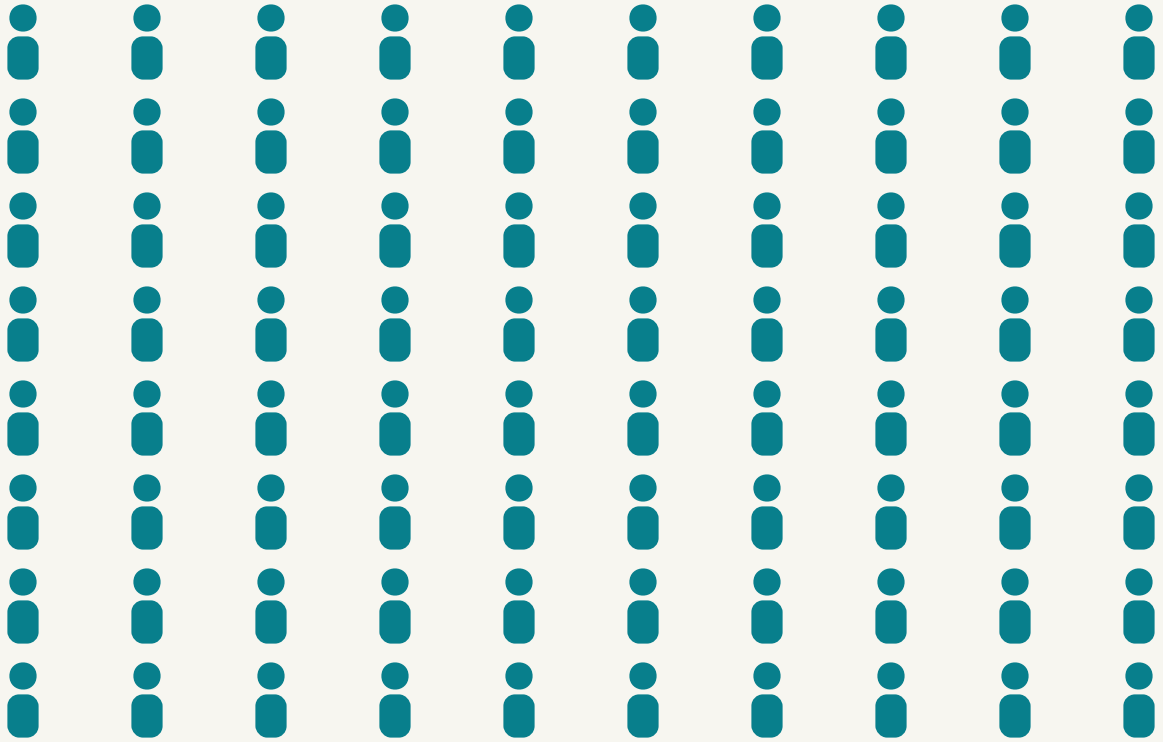
Sustained exposure

not one snapshot

Verify the exposure before explaining the mechanism.

Lim et al. 2024 · Discussion · Proposed measurements; no participant ketone values are invented.

Who stays in the picture?



Observed Same 80-person cohort

80

randomized at baseline

100% in the starting cohort

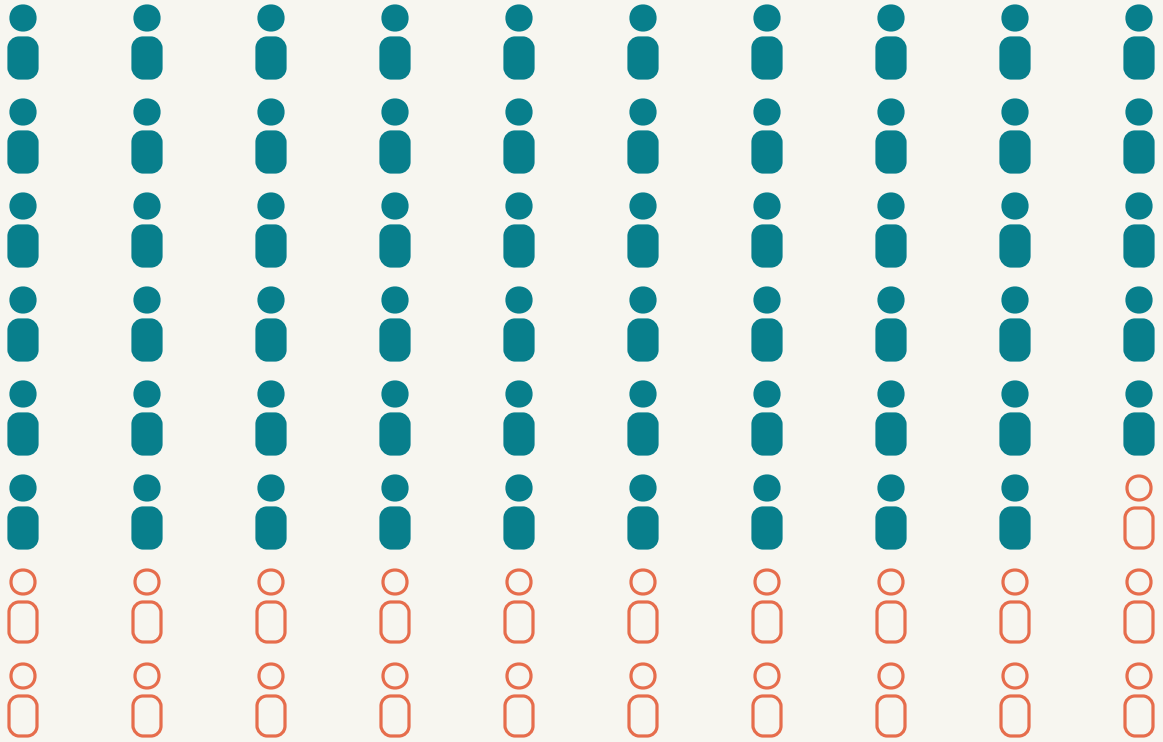
What happens next?

Follow the same cohort.

One figure = one participant

Lim et al. 2024 · Figure 1 + Table 2 · 6-month counts: HKD 29/41; ERD 30/39.

Who stays in the picture?



59

measured at 6 months

74% retained

21 outcomes unknown

Bias direction is uncertain.

At 12 months: 51 / 80 (64%)

Lim et al. 2024 · Figure 1 + Table 2 · 6-month counts: HKD 29/41; ERD 30/39.

Keep measuring, even after stopping.

Stop the diet?

Outcome collection

can still continue.



Keep measuring.

Record reasons.

Check who is missing.

ITT CLAIMED

Missing-outcome handling is unclear.

Retention + transparent reporting strengthen the comparison.

Lim et al. 2024 · Methods 2.9 + Figure 1 · Some exit reasons are reported; missing outcomes remain unknown.

One hospital is not everyone.

One hospital

Healthcare staff

86% women

English literacy + smartphone access

Singapore · selected adults with obesity



Other populations?

Applicability needs verification.

Not evidence of no benefit.

IMPROVEMENT: diverse, multicentre recruitment

Lim et al. 2024 · Methods 2.2 + Table 1 · 69 of 80 participants were women (86.25%).

A useful trial. A bounded conclusion.

WHAT THE DESIGN CAN EVALUATE

A supported dietary programme.

WHY it works	Mechanism remains uncertain.
WHO is missing	Follow-up can affect the comparison.
WHO it applies to	Broader applicability needs testing.

Useful evidence. Clear limits. Better next studies.

Source: Lim SL et al. (2024). Development and Pragmatic RCT of HKD Versus ERD. Nutrients 16:4380.